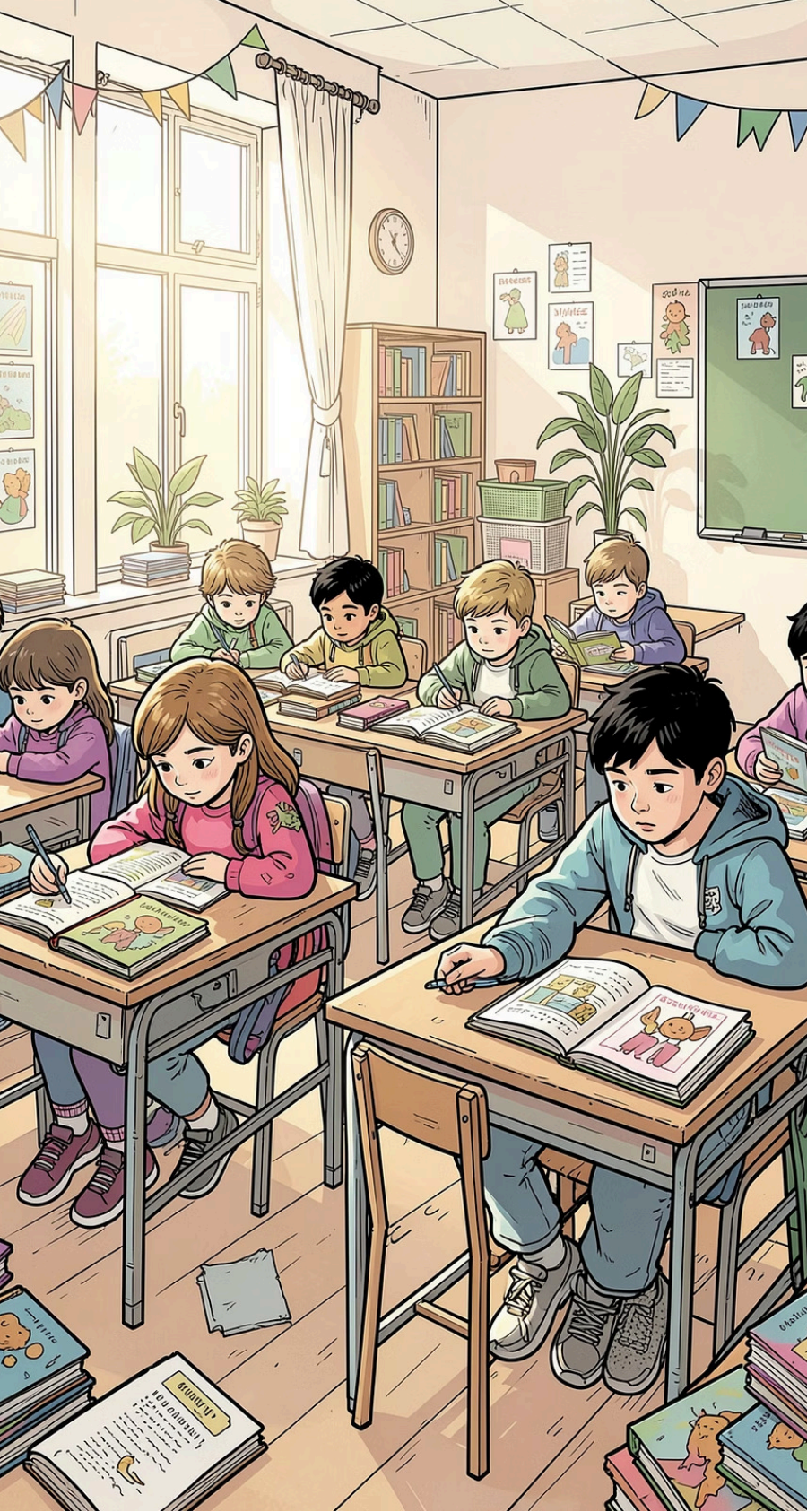


StoryWeaver

Building an AI Reading Co-Pilot for Early Readers





Reading Proficiency Is Becoming a National Crisis

31%

U.S. 4th Graders Proficient

Reading at or above grade level (NAEP)

69%

Below Proficiency

Elementary students in targeted regions

Students who fall behind in grades 1–3 rarely catch up. Teachers spend more time testing than teaching. The problem isn't decoding words — it's what happens next.



The real breakdown: Children lose comprehension → then confidence → then motivation. By the time adults notice, disengagement has already set in.

Why Existing Solutions Fall Short

What Apps Optimize For

Fluency drills

Gamification loops

Test score prep

What Stakeholders Actually Said

"It just looks like a bunch of words." — Teacher

"It's easier to just watch something." — Parent

Across interviews, a consistent pattern emerged: difficulty leads to frustration, frustration leads to avoidance, and avoidance becomes identity. **"I hate reading."**

Research Plan

Before designing a single feature, we set out to understand **why students disengage from reading** — and what adults around them actually experience.

Methods

01

30+ Interviews

Teachers, parents, reading specialists, tutors, and students

02

3 Ethnographic Observations

Live reading sessions in natural settings

03

5 Rounds of Usability Testing

Iterative validation with target users

04

Competitive Analysis

Landscape mapping of existing solutions

Frameworks Applied

Jobs To Be Done

5 Whys

Value vs. Effort

Affinity Mapping



Ethnographic Insights

Instead of only asking users what they do, **we watched**. Observing real reading sessions revealed behaviors that interviews never surfaced.

Children

- Skipped difficult words silently
- Pretended to understand
- Almost never asked for help
- Quietly disengaged over time

Parents

- Only learned about progress every few months
- Lacked confidence to help at home

Teachers

- Managed classrooms spanning multiple reading levels
- Relied heavily on delayed assessment data



Core Insight: Reading breakdown happens silently. By the time adults notice, the child has already checked out.

Three Personas, One Shared Problem

Research revealed three distinct users — each with different needs, but all affected by the same breakdown in early literacy support.



The Young Reader

Needs: Confidence, interesting content, help without embarrassment

"I don't understand."



The Overextended Parent

Needs: Visibility, conversation starters, simple progress reports

"I don't know how to help."



The Constrained Teacher

Needs: Scalable personalization, faster intervention, actionable analytics

"I don't have enough time."

The Silent Journey to Disengagement

Without timely support, a child's reading experience follows a predictable — and preventable — path. Each stage compounds the last, making intervention harder and rarer.



The opportunity: Intervene *before* frustration becomes identity. StoryWeaver is designed to break this cycle at the earliest possible moment.



How We Prioritized Features

Hundreds of ideas emerged from research. We scored each one against **user impact** and **engineering effort** to identify the highest-leverage opportunities.



Priority Tiers

Build Now – High Impact, Low Effort

Adaptive stories, vocabulary helper, parent & teacher dashboards

Build Next – Medium Priority

Summary questions, adaptive recommendations

Deprioritize – Low Value or High Cost

Pronunciation scoring, reading grading, heavy gamification



Product Strategy

Make struggling readers feel successful every day.



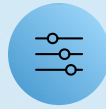
Personalized

Adapts to each reader's level and pace



Interest-Driven

Content matched to what kids actually care about



Adaptive

Difficulty adjusts in real time



Low-Friction

Supports adults without adding to their workload

MVP: Five Core Features

Each feature directly addresses the highest-value problems identified in research—built to intervene early, support adults, and keep kids reading.



Adaptive Reading

Content dynamically adjusts
Lexile level based on real-time
performance.



Vocabulary Assistant

Tap any word for a definition,
pronunciation, and usage
examples—instantly.



Reading Recommendation

Personalized suggestions
based on interests, reading
history, and difficulty.



Parent Dashboard

Progress trends, conversation
starters, and simple reports —
no guesswork.



Teacher Dashboard

Individual student progress,
intervention tools, and
actionable recommendations.

The Current Product

golden **STADIUM**. **Fans** wore bright red and blue **jerseys**. They cheered so loudly that the ground shook. Everyone waited for the **kickoff**. The game started with a long, high kick into the air. The **championship** had finally begun!

PART 2



By the third quarter, the score was tied. The star **QUARTERBACK** took a deep breath. He threw a perfect spiral down the field. A young receiver caught the ball with one hand! The crowd went wild. The runner was **swift** and dodged three **defenders**. He dove into the end zone for a spectacular touchdown.

let me try question 2 hmmm

 YouTube

StoryWeaver – Reading Co-Pilot Demo



 03:09

Product Decisions & Tradeoffs

We intentionally didn't build everything. These decisions reflect our research team's documented prioritization, balancing user impact and technical feasibility.

Live Pronunciation Grading

Rejected due to high cost, potential to increase student anxiety, and low confidence in its real-time accuracy.

Heavy Gamification

Rejected after teacher feedback indicated it often leads to compliance rather than genuine joy or intrinsic motivation.

AI Tutoring

Reduced in scope to focus on contextual vocabulary support, ensuring targeted help without overwhelming the user or system.

User Testing: Refining the Experience

Across five iterative testing rounds, we gathered invaluable feedback from students, parents, and teachers, directly informing product refinements.

What Resonated Most

Students Loved

- Choosing topics
- Vocabulary helper
- Adaptive difficulty

Parents Appreciated

- Conversation prompts
- Intuitive dashboards

Teachers Valued

- Intervention visibility
- Actionable insights

Key Product Improvements

Simplified Navigation

Streamlined flows reduced cognitive load for young readers and busy adults.

Reduced Dashboard Clutter

Focused data on what truly mattered to parents and teachers, eliminating noise.

Easier Recommendations

Enhanced the recommendation engine for more relevant and accessible content suggestions.

Improved Adaptive Suggestions

Fine-tuned the algorithm for more precise real-time difficulty adjustments, preventing frustration.

Results: From Research to Impact

Our comprehensive research approach led directly to the development of key features and measurable improvements for early readers.

Research Foundation

A deep dive into user needs and pain points established a solid base for product development.

- **30+** In-depth Interviews
- **3+** Ethnographic Studies
- **5+** Usability Testing Rounds

Key Product Outcomes

1

5+ Core Features Designed

- Adaptive Reading
- Vocabulary Support
- Teacher Dashboard
- Parent Dashboard
- Personalized Recommendations

2

Tangible Impact Delivered

- Improved Engagement
- Enhanced Comprehension
- Personalized Learning Experience

What Users Still Need

Even after refining the MVP, our users provided valuable feedback, highlighting areas where the co-pilot could further enhance their experience.

Parents: Accurate Assessment

Wanted assurance that progress reports genuinely reflected their child's true reading ability and progress, not just test scores.

Teachers: Real-time Struggle Insights

Needed immediate visibility into specific moments a student encountered difficulty, allowing for proactive, targeted intervention.

Students: Effortless Support

Expressed a desire for quick, unintrusive ways to get help when stuck, without feeling embarrassed or interrupting their reading flow.

Future Product Roadmap

Expanding StoryWeaver's capabilities to address deeper needs in assessment, classroom insights, and oral fluency.

AI Assessment Integrity

Problem: Parents and teachers need assurance that reading assessments accurately reflect a student's true ability.

Solution: Detects page skipping, rapid progression, or disengagement (e.g., looking away from screen) during assessments to flag potentially invalid sessions.

Outcome: An "Assessment Confidence Score" on dashboards increases trust in reading-level recommendations and progress tracking.

Intelligent Help Detection

Problem: Teachers often can't identify when students need help until issues become significant.

Solution: Computer vision detects raised hands during reading activities, logging help requests, wait times, and frequency. Dashboard insights highlight frequent/silent strugglers, average response times, and classroom support trends.

Impact: Turns previously invisible classroom behaviors into actionable insights for targeted educator and parent support.

Optional AI Pronunciation Coach

Problem: Some students struggle with pronunciation and oral fluency, needing low-pressure practice.

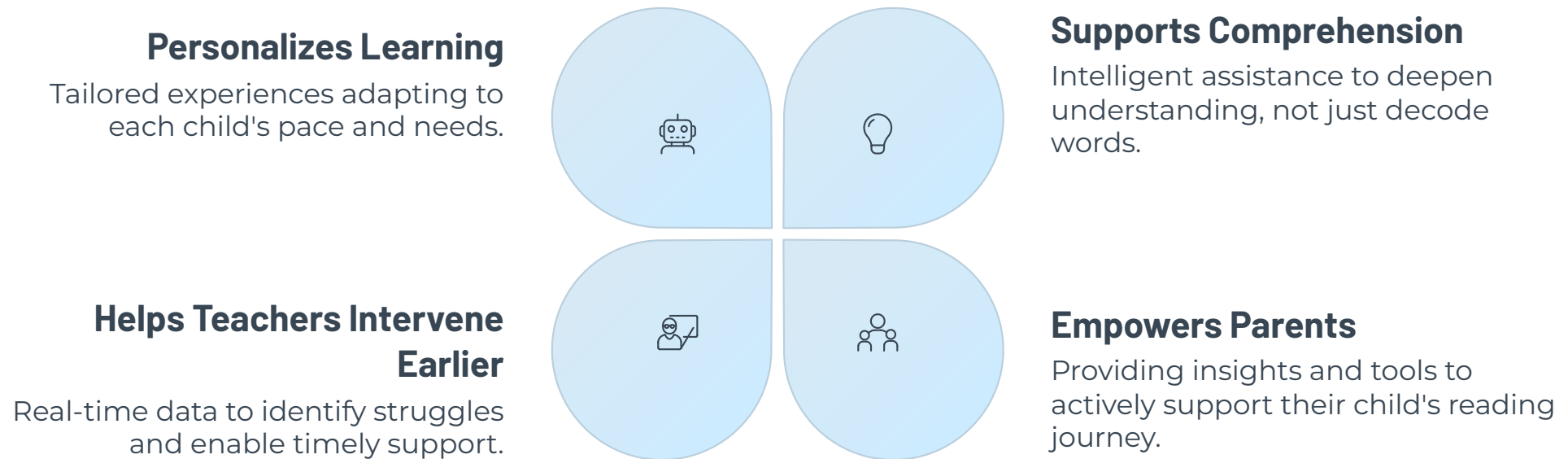
Solution: An optional AI-powered coach provides real-time feedback on pronunciation, pacing, and fluency, highlighting mispronounced words. Tracks improvement without affecting comprehension scores.

Why Optional: Prioritized as opt-in to avoid anxiety, building on the platform's core supportive experience.

Impact: Creates a holistic view of literacy by integrating comprehension, engagement, and oral reading fluency.

Long-Term Vision

StoryWeaver evolves beyond a simple reading app to become a comprehensive AI reading companion.



Our Core Goal

Catch frustration before it becomes disengagement.